

```
import java.io.BufferedReader;
import java.io.BufferedWriter;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;
import java.util.ArrayList;

public class GradeAnalyzer {
    private static int invalidLinesSkipped = 0;

    public static void main(String[] args) {
        ArrayList<Integer> scores = readScores("scores.txt");
        double average = calculateAverage(scores);

        int highest = Integer.MIN_VALUE;
        int lowest = Integer.MAX_VALUE;

        for (int score : scores) {
            if (score > highest) {
                highest = score;
            }
            if (score < lowest) {
                lowest = score;
            }
        }
    }
}
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    if (scores.isEmpty()) {
        System.out.println("No valid scores were found.");
    }

    writeReport(scores, average, highest, lowest, "report.txt");
}

// Returns a list of valid scores read from the file
public static ArrayList<Integer> readScores(String filename) {
    ArrayList<Integer> scores = new ArrayList<>();
    invalidLinesSkipped = 0;

    try (BufferedReader reader = new BufferedReader(new FileReader(filename))) {
        String line;

        while ((line = reader.readLine()) != null) {
            line = line.trim();

            if (line.isEmpty()) {
                invalidLinesSkipped++;
                System.out.println("Warning: blank line skipped.");
                continue;
            }

            try {
                int score = Integer.parseInt(line);

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        if (score < 0 || score > 100) {
            invalidLinesSkipped++;

            System.out.println("Warning: score out of range skipped: " + line);
        } else {
            scores.add(score);
        }
    } catch (NumberFormatException e) {
        invalidLinesSkipped++;

        System.out.println("Warning: invalid score skipped: " + line);
    }
}

} catch (IOException e) {
    System.out.println("Could not read file: " + e.getMessage());
}

return scores;
}

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// Returns the average of a list of scores, or 0.0 if the list is empty
public static double calculateAverage(ArrayList<Integer> scores) {
    if (scores.isEmpty()) {
        return 0.0;
    }
}

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double total = 0.0;

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    for (int score : scores) {
        total += score;
    }

    return total / scores.size();
}

// Writes and prints the report
public static void writeReport(ArrayList<Integer> scores,
                                double avg, int high, int low,
                                String outputFile) {

    int countA = 0;
    int countB = 0;
    int countC = 0;
    int countD = 0;
    int countF = 0;

    for (int score : scores) {
        if (score >= 90) {
            countA++;
        } else if (score >= 80) {
            countB++;
        } else if (score >= 70) {
            countC++;
        } else if (score >= 60) {
            countD++;
        }
    }
}

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    } else {  
        countF++;  
    }  
}
```

```
String highestText = scores.isEmpty() ? "N/A" : Integer.toString(high);
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String lowestText = scores.isEmpty() ? "N/A" : Integer.toString(low);
```

```
String report = String.format(  
    "=== Grade Analysis Report ===%n"
```

```
    + "Total scores processed: %d%n"  
    + "Invalid lines skipped: %d%n%n"  
    + "Average score: %.2f%n"  
    + "Highest score: %s%n"  
    + "Lowest score: %s%n%n"  
    + "Grade distribution:%n"  
    + " A (90-100): %d%n"  
    + " B (80-89): %d%n"  
    + " C (70-79): %d%n"  
    + " D (60-69): %d%n"  
    + " F (below 60): %d%n",
```

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scores.size(), invalidLinesSkipped, avg, highestText, lowestText,
```

```
countA, countB, countC, countD, countF);
```

```
System.out.print(report);
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```
try (BufferedWriter writer = new BufferedWriter(new FileWriter(outputFile))) {
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```
        writer.write(report);
    } catch (IOException e) {
        System.out.println("Could not write report: " + e.getMessage());
    }
}
}
```